

Seventh International Olympiad, 1965

1965/1

Determine all values x in the interval $0 \leq x \leq 2\pi$ which satisfy the inequality

$$2 \cos x \leq \sqrt{1 + \sin 2x} - \sqrt{1 - \sin 2x} < \sqrt{2}.$$

1965/2

Consider the system of equations

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = 0,$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = 0,$$

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = 0,$$

with unknowns x_1, x_2, x_3 . The coefficients satisfy the conditions:

[(a)] a_{11}, a_{22}, a_{33} are positive numbers;

[(b)] the remaining coefficients are negative numbers;

[(c)] in each equation, the sum of the coefficients is positive.

Prove that the given system has only the solution $x_1 = x_2 = x_3 = 0$.

1965/3

Given the tetrahedron $ABCD$ whose edges AB and CD have lengths a and b respectively. The distance between the skew lines AB and CD is d , and the angle between them is ω . Tetrahedron $ABCD$ is divided into two solids by plane ε , parallel to lines AB and CD . The ratio of the distances of ε from AB and CD is equal to k . Compute the ratio of the volumes of the two solids obtained.

1965/4

Find all sets of four real numbers x_1, x_2, x_3, x_4 such that the sum of any one and the product of the other three is equal to 2.

1965/5

Consider $\triangle OAB$ with acute angle AOB . Through a point $M \neq O$ perpendiculars are drawn to OA and OB , the feet of which are P and Q respectively. The point of intersection of the altitudes of $\triangle OPQ$ is H . What is the locus of H if M is permitted to range over:

[(a)] the side AB ,

[(b)] the interior of $\triangle OAB$?